University of Liège

Project presented as part of the Climate and Environmental Modeling (SPAT0025-1) course

Development of an Energy Balance Model and analysis of climate sensitivity to the solar constant

Louis CLEEN

Academic year 2025–2026

Contents

Introduction **2**
Context 2
Objectives 2
Outline 3

1 Methodology **4**
1.1 Theoretical framework 4
1.1.1 Annual mean incident solar flux 4
1.1.2 Seasonal incident solar flux 5
1.1.3 Albedo parameterization 7
1.1.4 Outgoing radiative flux (infrared) 8
1.1.5 Meridional heat transport - Budyko's formulation 8

2 Results **10**
2.1 Overview 10
2.2 Entropy production 11
2.3 Sensitivity to the solar constant 12

Introduction

Context

Earth's climate results from complex interactions between the atmosphere, the oceans, the cryosphere, and the land surface. To better understand and predict how it evolves, scientists have sought to describe the climate system based on fundamental physical principles, such as the balance between the solar energy received and the thermal radiation emitted by the planet, as described by Stefan-Boltzmann's law.

Given that the scale of these phenomena makes experimental work in the laboratory almost impossible, numerical models have thus proven to be valuable tools for simulating the behavior of the climate system as a function of various physical and chemical parameters.

It is within this framework that the first Energy Balance Models (EBMs) were developed. These simple models made it possible to relate the Earth's average temperature to key parameters such as albedo or atmospheric composition, and they constituted an essential step in understanding the mechanisms of the climate.

With the increase in computational power and the accumulation of knowledge about the physical processes involved, more complex climate models have gradually been developed. Today's three-dimensional General Circulation Models (GCMs), used in particular by the IPCC, incorporate many coupled phenomena (atmospheric and oceanic circulation, the water cycle, cloud interactions, etc.), enabling detailed simulations of the climate system. However, this complexity makes their interpretation more challenging and their computational cost very high.

Despite these advances, EBMs continue to play a fundamental role. Their simplicity makes it possible to isolate and analyze the essential mechanisms of the climate, while providing an accessible framework for testing hypotheses and exploring the climate system's sensitivity to various parameters. One-dimensional models (1D EBMs) introduce a spatial dependence, typically latitude, while the vertical component is neglected. This is sufficient to represent temperature variations between the equator and the poles, and they therefore represent a relevant compromise between simplicity and realism.

Among the major contributions to the development of this type of model, the work of Budyko on meridional heat transport plays a central place. In 1969, he proposed parameterizing this flux as a function of the temperature difference between adjacent latitude bands. As we will see, this approximation leads to a simple numerical implementation and yields results in good agreement with observations.

Objectives

The objective of this project is to develop a one-dimensional energy balance model using Budyko's (1969) formulation to describe meridional heat transport. To this end, the

Earth will be divided into latitudinal bands. Specifically, 18 bands, each 10° wide, will be considered here. Then, an analysis of climate sensitivity to the solar constant will be conducted.

Outline

The first chapter will be devoted to a theoretical description of the model and its implementation. In particular, the various terms involved in its main equation will be examined. Two implementations will be presented for the incident solar flux: the former will simply consist of an annual average, and the latter will take seasonality into account. The second chapter will be dedicated to the results and the impact of the solar constant on the climate.

Chapter 1

Methodology

1.1 Theoretical framework

In a 1D energy balance model (EBM), the variable of interest is the surface temperature T, which depends on latitude and time. The partial differential equation governing its evolution can be written as follows:

$$C_m \frac{\partial T(\phi, t)}{\partial t} = R_{\downarrow}(\phi, t) - R_{\uparrow}(\phi) + \text{ net transport into zone } \phi$$

where:

- C_m is the heat capacity of the surface and the atmosphere in $\mathrm{J\,m^{-2}\,K^{-1}}$. It plays a role of thermal inertia. In theory, it depends on the latitude band considered, but it is often assumed to be constant for the sake of simplicity.
- $R_{\downarrow}(\phi)$ is the absorbed incident solar radiative flux in $\mathrm{W\,m^{-2}}$. In the case of an annually averaged model, this quantity does not depend on time.
- $R_{\uparrow}(\phi)$ is the outgoing radiative flux (infrared) at the top of the atmosphere in $\mathrm{W\,m^{-2}}$.
- net transport into zone ϕ related to the meridional transport of heat along latitude in $\mathrm{W\,m^{-2}}$.

It should be noted that the hydrological cycle is not represented and, therefore cloud cover should be impose.

By discretizing by latitude band i, it can be rewritten as

$$C_i \frac{\partial T_i}{\partial t} = S_i(1 - \alpha(T_i)) - R_{\uparrow}(T_i) - F(T_i) \tag{1.1}$$

where α is the albedo, S_i the solar flux and $F(T_i)$ the meridional heat transfer

1.1.1 Annual mean incident solar flux

Incident solar flux refers to the radiative flux per unit area received at the top of the Earth's atmosphere. This quantity constitutes the primary source of energy for the climate system.

Its value, known as the "solar constant" S_0 is approximately 1361 W/m^2. When averaged over the entire Earth's surface, this flux is reduced by a factor of one-fourth[1].

In the case of an annual mean flux, there is no temporal dependence, and we can write for a band i,

$$S_i = \frac{S_0}{4} s(x_i)$$

where $s(x_i)$ a function of latitude, normalized such that for n bands we have : $\sum_i^n s(x_i)\, f_i = 1$ with f_i the area fraction of band i.

One way to calculate $s(x_i)$ is to use a Legendre polynomial series expansion. These polynomials are well-suited for approximating a function on a spherical geometry. Assuming symmetry between the northern and southern hemispheres, only the even-order terms remain: P_0, P_2, P_4, etc. Truncating the series after the first non-trivial term, we can have, $s(x_i) \approx a_0 P_0(x_i) + a_2 P_2(x_i)$ with $x_i = \sin\phi_i$. Since $\forall x$, $P_0(x) = 1$, we can write: $S(\phi_i) = \frac{S}{4} a_0 \left[1 + \frac{a_2}{a_0} P_2(\sin\phi_i)\right]$ which leads to

$$S_i = \frac{S_0}{4}\left[1 + s_2 P_2(\sin\phi_i)\right]$$

in order to satisfy the normality condition.

Finally, by the definition of P_2 and by a numerical adjustment that yields to $s_2 \approx -0.482$, we have

$$S_i = \frac{S_0}{4}\left[1 - \frac{0.482}{2}\left(3\sin^2\phi_i - 1\right)\right]$$

It should be noted that since this function is time-independent, it can be precomputed for each latitude band once and for all.

1.1.2 Seasonal incident solar flux

In the case of a seasonal model, the solar irradiance received depends of the time and the amount received on a horizontal surface at the top of the atmosphere is calculated using a projection that varies with h, the sun's elevation:

$$S = S_0 \left(\frac{d_0}{d_s}\right)^2 \sin h$$

where $\left(\frac{d_0}{d_s}\right)^2$ is a factor that corrects the intensity based on the Earth-Sun distance d_s (which fluctuates from around its reference value d_0 due to the eccentricity of the orbit). The exponent is justified by the intensity of the flux, which is emitted isotropically and distributed over the surface of a sphere, whose area increases in proportion to the square of the radius.

Thanks to Kepler's theory, this factor can be written as

$$\left(\frac{d_0}{d_s(t)}\right)^2 = \frac{1}{(1 - e\cos E)^2}$$

with e the orbital eccentricity, E the eccentric anomaly

[1] As seen from the Sun, the Earth intercepts radiation over a disk of area πR^2. However, due to its spherical geometry, the energy received is spread over a surface area of $4\pi R^2$. Therefore, the resulting mean flux is: $\frac{\pi R^2}{4\pi R^2} S_0 = \frac{S_0}{4}$

For small eccentricity ($e \ll 1$), a first-order Taylor expansion yields

$$\left(\frac{d_0}{d_s(t)}\right)^2 \approx 1 + 2e\cos E$$

Since the eccentricity is small, the seasonal variation can be approximated by a cosine function of time. We therefore write

$$\left(\frac{d_0}{d_s(t)}\right)^2 \approx 1 + 2e\cos(\frac{2\pi}{Y}(t - t_p))$$

where t_p is chosen such that the maximum occurs at perihelion (around January 3) and with Y the periodicity, which is one year here.

Since, $h = \frac{\pi}{2} - \vartheta_0$ with ϑ_0 the solar zenith angle, we have $\sin h = \cos\vartheta_o$. Using a spherical trigonometry formula, we have

$$\cos\theta_o = \sin\delta\sin\phi + \cos\delta\cos\phi\cos AH$$

where δ is the solar declination and AH the true hour angle.

Solar declination

The declination δ is a function of time that depends on the obliquity ϵ. This obliquity oscillates slightly over time (Milankovitch cycles), but assuming it is constant, we obtain the following projection:

$$\delta(t) = \arcsin(\sin\epsilon\ \sin\lambda(t))$$

By making the approximation of a circular orbit (constant angular velocity), we can write:

$$\lambda(t) = \frac{2\pi}{Y}(t - t_v)$$

where $t = 0$ corresponds to January 1, t_v to the day of the vernal (spring) equinox in the Northern Hemisphere (around March 21).

True hour angle

It is not necessary to know the value of the hour angle at all times, as in practice we use a large integration step (> 1 day), which is sufficiently accurate. Instead, we'll do a one-day integration as follows,

$$Q(\phi, t) \stackrel{\text{def}}{=} \int_0^{24\text{h}} S_0\cos\vartheta_0(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_0\max(0, \cos\vartheta_0)\,\mathrm{d}AH$$

However, the sun is visible only when $-H_0 \leq AH \leq H_0$ where H_0 denotes the solar hour angle at sunset. So we have,

$$\begin{aligned} Q(\phi, t) &= \frac{1}{\pi}\int_0^{H_0} S_0\cos\vartheta_0\,\mathrm{d}AH \\ &= \frac{S_0}{\pi}\left(H_0\sin\phi\sin\delta + \cos\phi\cos\delta\sin H_0\right) \end{aligned}$$

At sunset, the solar zenith angle is equal to 90° which leads to

$$\sin\delta\sin\phi + \cos\delta\cos\phi\cos H_0 = 0$$

and implies

$$\cos H_0 = -\tan\delta\ \tan\phi$$

1.1.3 Albedo parameterization

Albedo is the proportion of light that a surface reflects. For each latitude band, it depends on the type of surface (land or ocean), the presence of snow or sea ice, and cloud cover. We therefore understand that it varies with temperature (i.e., the ice-albedo feedback). As a good approximation, we will use a parameterization that consists of averaging the albedo of clear-sky and cloudy-sky regions:

$$\alpha_i = f_{\text{cloud}}\, \alpha_{\text{cloudy_sky}} + (1 - f_{\text{cloud}})\, \alpha_{\text{clear_sky}}$$

- where $\alpha_{\text{clear_sky}}$ combines the albedo of the atmosphere (backscattering) and that of the surface. Figure 1.1 illustrates the calculation, which takes the form of a multiple reflection series, of which only the first few terms contribute significantly. We will retain:

$$\alpha_{\text{clear_sky}} = \alpha_{\text{atm}} + (1 - \alpha_{\text{atm}})^2\, \alpha_{\text{surf}}$$

- and where $\alpha_{\text{cloudy_sky}}$ will be taken as,

$$\alpha_{\text{cloudy_sky}} = \frac{\alpha_{\text{atm}}}{2} + \left(1 - \frac{\alpha_{\text{atm}}}{2}\right)^2 \alpha_{\text{cloud}}$$

  because the tops of the clouds are located higher up in the atmosphere, so we arbitrarily divide by 2.

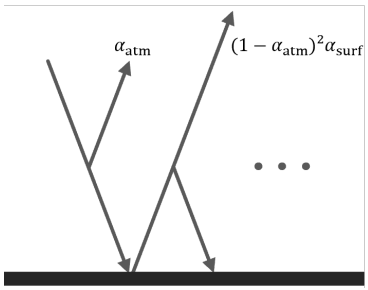

Figure 1.1: Schematic of radiative reflections between atmosphere and surface, showing contributions of atmospheric and surface albedo

Since the albedo of the atmosphere and clouds is considered constant, all that remains is to express the surface albedo. We will use the following formulation:

$$\alpha_{\text{surf}} = f_{\text{land}}\, (f_{\text{snow}}\, \alpha_{\text{snow}} + (1 - f_{\text{snow}})\, \alpha_{\text{land}}) + (1 - f_{\text{land}})(f_{\text{ice}}\, \alpha_{\text{ice}} + (1 - f_{\text{ice}})\, \alpha_{\text{ocean}})$$

where the fractions of sea ice f_{ice} and continental snow/ice f_{snow} will be parameterized through various critical thresholds and linearly with temperature:

$$f_{\text{ice}} = \begin{cases} 1 & \text{if } T_i < T_{\text{inf,ice}} \\ \frac{T_{\text{sup,ice}} - T_i}{T_{\text{sup,ice}} - T_{\text{inf,ice}}} & \text{if } T_{\text{inf,ice}} < T_i < T_{\text{sup,ice}} \\ 0 & \text{if } T_i > T_{\text{sup,ice}} \end{cases}, \quad f_{\text{snow}} = \begin{cases} 1 & \text{if } T_i < T_{\text{inf,snow}} \\ \frac{T_{\text{sup,snow}} - T_i}{T_{\text{sup,snow}} - T_{\text{inf,snow}}} & \text{if } T_{\text{inf,snow}} < T_i < T_{\text{sup,snow}} \\ 0 & \text{if } T_i > T_{\text{sup,snow}} \end{cases}$$

1.1.4 Outgoing radiative flux (infrared)

Outgoing infrared radiation is a central component of energy balance models. It corresponds to the thermal energy emitted by the Earth's surface and the atmosphere into space in the form of infrared radiation.

According to the blackbody radiation law formulated by Stefan-Boltzmann, this flux depends strongly on temperature: the hotter the Earth is, the more energy it emits, in proportion to its surface temperature. We can write:

$$R_{\uparrow} = \tau \epsilon \sigma T_i^4$$

where, ϵ is the effective emissivity (≈ 1 for a blackbody), σ is the Stefan-Boltzmann constant, and τ is the fraction of flux escaping into space, in other words, the atmospheric transmissivity factor. For example, this factor can be described empirically as $1 - m_i \tanh(19\, T_i^6 \times 10^{-16})$, with m_i the atmospheric opacity.

However, this model quickly reaches its limits because it does not explicitly account for anthropogenic greenhouse gas emissions. These emissions absorb some of the infrared radiation emitted by the surface and re-emit it in all directions, including back toward the ground. To account for this effect, we will use the following approximation:

$$R_{\uparrow}(T_i) = (A + B\, T_i(^{\circ}\mathrm{C})) - (C + D\, T_i(^{\circ}\mathrm{C}))\ f_{\mathrm{cloud}}$$

with: $A = 243.39 - 4.48 \ln\left(\frac{\mathrm{pCO2}}{\mathrm{pCO2}_0}\right)$, $B = 2.07 - 0.0514 \ln\left(\frac{\mathrm{pCO2}}{\mathrm{pCO2}_0}\right)$, $C = 54.13$ and $D = 0.58$.

It is based on the linearization of the radiation emitted in clear-sky conditions, $A + B\, T_i$, where A and B depend on the ratio of the current partial pressure (or concentration) of CO_2 (at time t) to the pre-industrial level (approximately 280 ppm). As the concentration of CO2 increases, the quantities A and B decrease, leading to a decrease in outgoing radiative flux and thus global warming, which is consistent with the theory.

The term $-(C + D\, T_i(^{\circ}\mathrm{C}))\ f_{\mathrm{cloud}}$ is used to account for cloud cover, which tends to reduce the outgoing infrared flux. Indeed, the flux absorbed by the clouds is re-emitted in all directions, and thus some of it returns to the ground.

1.1.5 Meridional heat transport - Budyko's formulation

The meridional heat transport refers to the displacement of thermal energy from equatorial regions toward the poles. This process is essential for regulating Earth's climate, as it compensates for the imbalance in solar radiation between low and high latitudes.

It is primarily driven by two major systems: the atmosphere (through winds and circulations such as Hadley cells) and the oceans (through currents as the Gulf Stream). Together, they redistribute excess heat from the tropics to colder regions, thereby reducing extreme temperature differences across the planet. This transport plays a key role in global climate dynamics, and its variations can influence phenomena such as climate change, precipitation patterns, and extreme weather events.

Unlike GCMs, where this flux is explicitly simulated, EBMs adopt a simplified approach by parameterizing it. Budyko's formulation is one of the simplest in this regard and consists of representing meridional transport (more precisely, the divergence of its flux) as a relaxation toward a global mean temperature:

$$F(T_i) = \mathrm{div}(F_{mer}) = k_t(T_i - \overline{T}) \tag{1.2}$$

where k_t is an empirically determined constant in $\mathrm{W\,m^{-2}\,K^{-1}}$

Since k_t is independent of latitude, we can write the following energy conservation equation:

$$\sum_{i=1}^{n} A_i F(T_i) = k_t \sum_{i=1}^{n} A_i (T_i - \overline{T}) = \sum_{i=1}^{n} A_i T_i - \overline{T} \sum_{i=1}^{n} A_i = 0 \tag{1.3}$$

where A_i is the area of latitude band i.

Chapter 2

Results

2.1 Overview

Let's begin with some general results to examine the model's behavior and verify its convergence.

The solver used to integrate the model is an explicit Euler method. This method proved to be sufficient and stable enough to allow the model to converge to an equilibrium state. As a result, it was not considered necessary to implement more complex methods. In the case of a seasonal model (see left plot of Figure 2.1), this equilibrium is dynamic, or more precisely cyclic. The average planetary temperature is approximately 12.18 °C.

The right side shows the equilibrium temperature profile for both model versions[1], under pre-industrial conditions. It can be seen that the two models converge well towards a similar result.

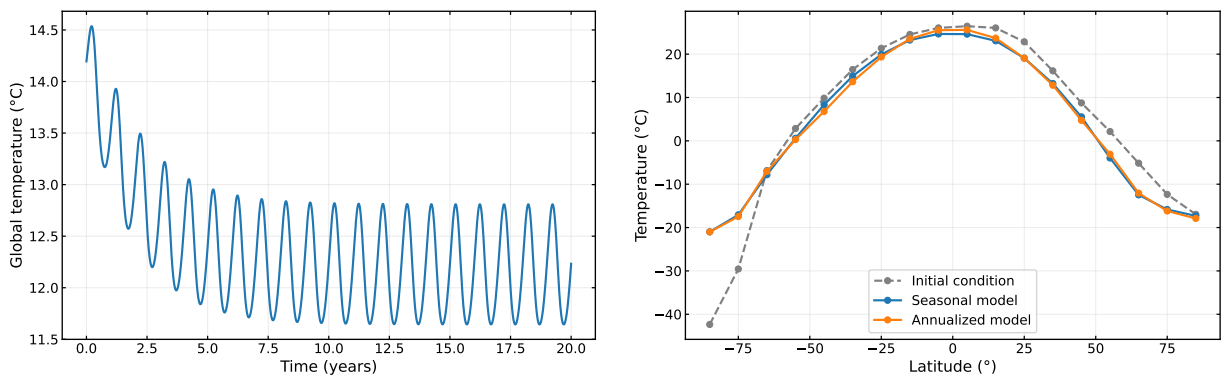

Figure 2.1: (Left) Global temperature history for a seasonal EBM. (Right) Latitudinal temperature profiles at equilibrium.

Figure 2.2 shows the temperature history over the last two years integration (at equilibrium) for the seasonal model's. The higher the latitude, the greater the temperature variations throughout the year, which is consistent with reality. A second observation is that the system responds with some latency (remember that the summer solstice is on June 21 in the Northern Hemisphere). This is mainly due to the heat capacity, which can be adjusted in the configuration file. A low value results in a more responsive system, while a higher value simulates a system that mixes a deeper layer of the ocean.

[1] Averaged over the last year for the seasonal case

The heatmap on the right-hand side of Figure 2.2 shows solar flux throughout the year. It gives a clearer visualization of the dependence on latitude and time. As it can be seen, the flux is fairly constant near the equator. In fact, at this latitude, the duration of the day is around 12 hours during the whole year. One can now better understand the small variation in temperature shown in the temperature history. The flux reaches its maximum at the poles during the summer solstice (around June 21 in the Northern Hemisphere and December 21 in the Southern Hemisphere). Thus, somewhat surprisingly, the flux at these times is higher than at the equator. However, it should not be forgotten that the presence of ice and snow at these latitudes increases the albedo and therefore reduces the amount of flux actually absorbed.

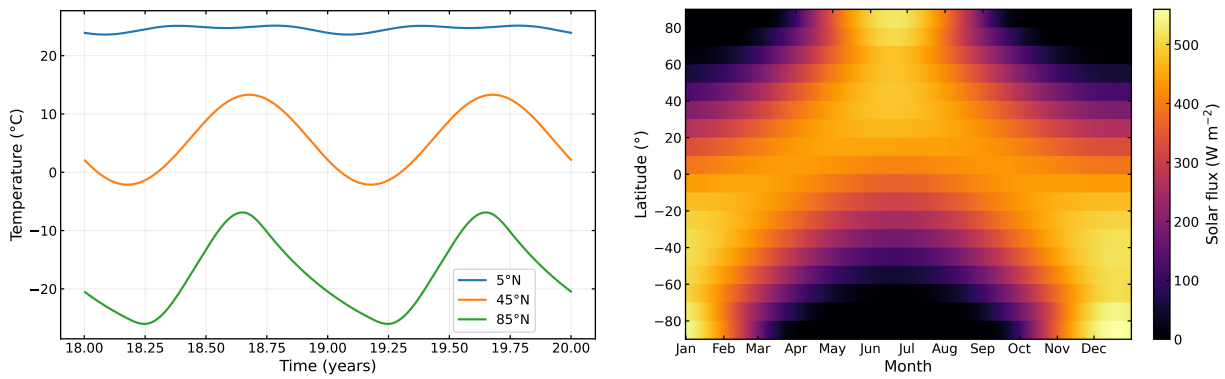

Figure 2.2: (Left) Temperature history over the last two years at different latitudes. (Right) Heatmap of solar flux throughout the year.

2.2 Entropy production

In 1975, Garth Paltridge showed that the meridional heat transport on present-day Earth was such that it corresponded to a maximum in entropy production σ. This idea is useful because it can be applied to determine the optimal meridional heat transport coefficient in EBMs.

Entropy production can be written as follows:

$$\sigma = \sum_i A_i \frac{\delta \dot{Q}_i}{T_i} = \sum_{i=1}^{n} A_i F_i \left(\frac{1}{\bar{T}} - \frac{1}{T_i} \right) \tag{2.1}$$

$$= k_t \sum_{i=1}^{n} A_i \frac{(T_i - \bar{T})^2}{T_i \bar{T}}$$

This shows that we indeed have a function that is always positive. Thanks to energy conservation (1.3), we can rewrite (2.1) in a way that is faster to compute:

$$\sigma = k_t \sum_{i=1}^{n} A_i \frac{\bar{T} - T_i}{T_i}$$

In practice, the script `run_entropy_production.py` integrates the model over a range of k_t values and computes the entropy production for each case. The graph on the left in Figure 2.3 shows the effect on temperature for a selection of values. One can see that as k_t increases, temperatures at high latitudes also rise, which can be explained by more

efficient heat transport. The right-hand side displays the entropy production values. The maximum is reached for $kt = 1.3$. However, at this value, Earth's global mean temperature would be lower than expected. It is important to keep in mind that Paltridge's hypothesis is not perfect: the system stabilizes near the maximum, but not exactly at it.

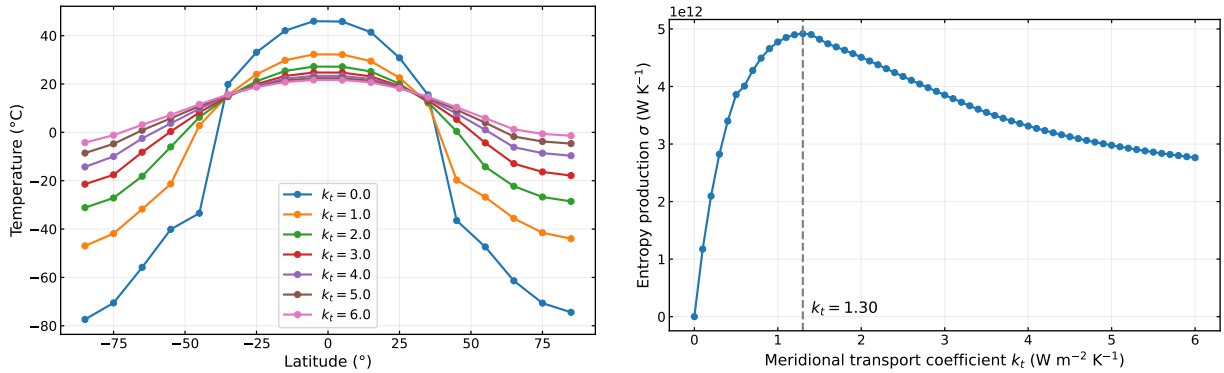

Figure 2.3: (Left) Latitudinal temperature profiles for different k_t values. (Right) Entropy production as a function of k_t.

2.3 Sensitivity to the solar constant

As a starting point for analyzing the climate's response to the solar constant, the effect of a 5% increase is considered. It seems obvious that, in this situation, the climate will warm, but it may be interesting to examine how the proportion of sea ice evolves over the course of the year. Figure 2.4 shows that for the +5% scenario, ice is no longer maintained year-round at the poles. We can see that it disappears from July until mid-October in the Northern Hemisphere. This shift is mainly related to thermal inertia, but also to latent heat (which is not explicitly modelled). It should be noted that in the simulation carried out here, the first latitude band (-90° to -80°) is treated as being exclusively continental. As a result, no sea ice can form there (instead the model uses continental snow for the albedo).

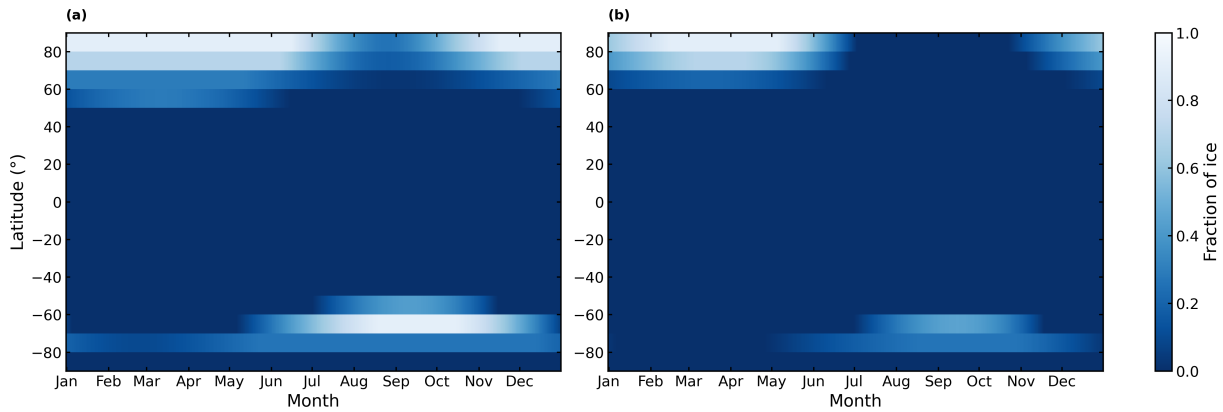

Figure 2.4: Distribution of sea ice fraction. (a) Scenario of reference for $S_0 = 1368$ W m^{-2}. (b) Situation where S_0 is increased by 5%.

To continue this study, we can examine the climate sensitivity over a range of values of the solar constant. As shown in Figure 2.5, the response is not purely linear and exhibits an

abrupt jump. This effect is related to the ice–albedo feedback. To illustrate this, the image on the right shows the evolution of the albedo for different values of the solar constant.

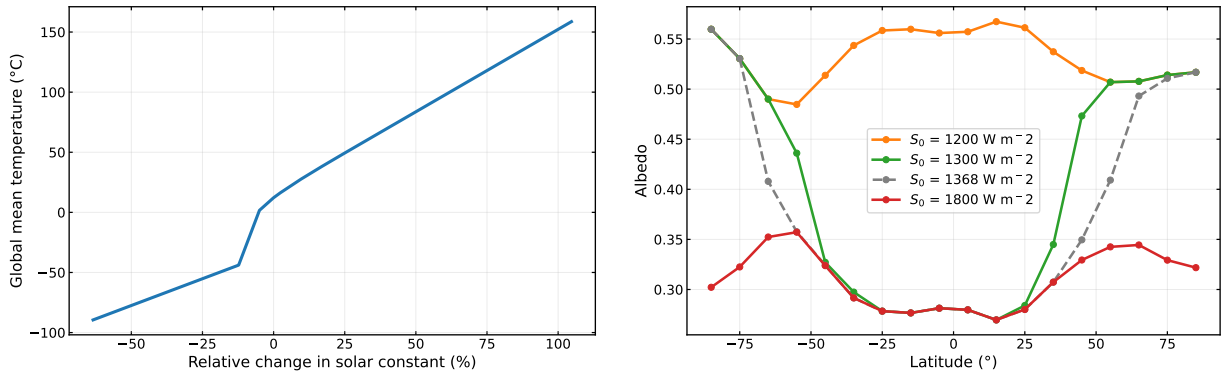

Figure 2.5: (Left) Global mean temperature sensitivity to solar constant changes. (Right) Latitudinal albedo profile at equilibrium.

To better quantify the ice melt described previoulsy, an estimate of the ice extent was carried out for both hemispheres. To do this, the minimum and maximum latitudes of sea ice over the course of a year are plotted as a function of the solar constant. The left-hand graph in Figure 2.6 shows the result. It can be noted that the model is capable of reproducing the "Snowball Earth" hypothesis, according to which the entire Earth was covered with ice about 700 million years ago. However, it should be pointed out that a decrease of about 7% in the solar constant (associated with this period) is not sufficient. This is why the hypothesis is instead linked to a fall in atmospheric CO_2. The graph on the right shows the evolution of the Earth's global albedo as a function of S_0. This provides an even clearer visualization of the transition to the Snowball Earth state, as well as the ice–albedo feedback.

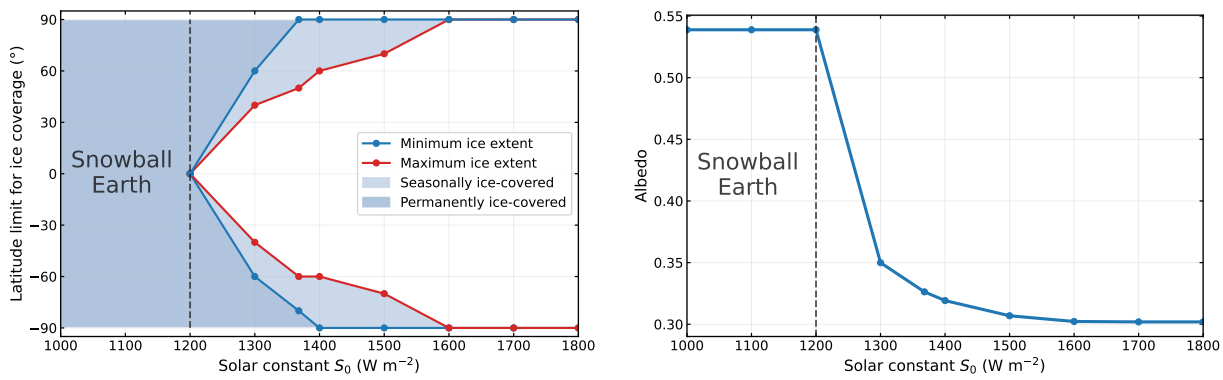

Figure 2.6: (Left) Sea ice extent as a function of solar the constant. (Right) Global mean albedo as a function of the solar constant.